

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Heredity and Evolution

Formative-only topic in CBSE 2026-27: useful for school assessment; not year-end board assessed

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

In a cross between pure tall and pure dwarf pea plants, what appears in F1 and F2 generations?

Answer:

If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Self-crossing $Tt \times Tt$ gives F2 genotypes 1 TT:2 Tt:1 tt and phenotype ratio 3 tall:1 dwarf.

Question 2

How is the sex of a human child determined?

Answer:

The mother contributes an X chromosome in the egg. The father's sperm contributes X or Y: XX typically develops as female and XY as male. The sperm therefore determines the chromosomal combination.

Question 3

Distinguish inherited traits from acquired traits.

Answer:

Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Acquired traits develop during life, such as a learned skill, and usually are not passed through DNA.

Question 4

What is the central idea or skill tested in Heredity and Evolution?

Answer:

The central focus is: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Heredity and Evolution.

Answer:

First point: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Second point: The mother contributes an X chromosome in the egg. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Heredity and Evolution lead to an important result or conclusion?

Answer:

Begin with the key idea: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Then connect it to the result: The mother contributes an X chromosome in the egg.

Heredity and Evolution - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Heredity and Evolution?

Answer:

Use this supporting detail: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Self-crossing Tt x Tt gives F2 genotypes 1 TT:2 Tt:1 tt and phenotype ratio 3 tall:1 dwarf. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Heredity and Evolution.

Answer:

The mother contributes an X chromosome in the egg. The father's sperm contributes X or Y: XX typically develops as female and XY as male. The sperm therefore determines the chromosomal combination. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Heredity and Evolution?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour.

Question 10

Compare the first and second important ideas in Heredity and Evolution.

Answer:

The first idea is: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. The second idea is: The mother contributes an X chromosome in the egg. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Heredity and Evolution.

Answer:

Cause or starting point: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Effect or outcome: The mother contributes an X chromosome in the egg. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Heredity and Evolution?

Answer:

A clear sequence is: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Next, The mother contributes an X chromosome in the egg. Finally, Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour.

Heredity and Evolution - Continued

Question 13

Give a 40-50 word answer outline for an important question from Heredity and Evolution.

Answer:

Start with the direct answer: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Add one supporting point: The mother contributes an X chromosome in the egg. End with the outcome or message: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Heredity and Evolution?

Answer:

Use an accurate scientific statement: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Heredity and Evolution?

Answer:

Connect the idea to its reason or result. For instance: The mother contributes an X chromosome in the egg. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Heredity and Evolution.

Answer:

A sound inference is that inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Heredity and Evolution.

Answer:

Apply the concept by first stating: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Then justify the response with: The mother contributes an X chromosome in the egg.

Question 18

What would make an answer from Heredity and Evolution incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Also add: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour.

Heredity and Evolution - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Heredity and Evolution.

Answer:

Assertion: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Reason: The mother contributes an X chromosome in the egg. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Heredity and Evolution in two clear steps?

Answer:

Step 1: state the concept or event - If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Step 2: support it with the explanation or result - The mother contributes an X chromosome in the egg.

Question 21

What answer structure is suitable for a 3-mark question from Heredity and Evolution?

Answer:

Use three connected points: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. The mother contributes an X chromosome in the egg. Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Heredity and Evolution?

Answer:

Give an introduction, develop these key points - If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. The mother contributes an X chromosome in the egg. - and close with: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Heredity and Evolution be applied to a new situation or case study?

Answer:

Identify the relevant principle first: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Then use the chapter's reasoning to explain the case: The mother contributes an X chromosome in the egg.

Question 24

Which detail from Heredity and Evolution is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on Heredity and Evolution.

Answer:

A suitable conclusion is: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. This conclusion should follow naturally from the explanation rather than introduce a new point.

Heredity and Evolution - Continued

Question 26

Make a three-point revision note for Heredity and Evolution.

Answer:

1. If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. 2. The mother contributes an X chromosome in the egg. 3. Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Heredity and Evolution before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. - and supported it with a reason, example, calculation, or outcome such as: The mother contributes an X chromosome in the egg.

Question 28

Why does Heredity and Evolution remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. The mother contributes an X chromosome in the egg. Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour.

Question 29

Create a final 60-80 word answer plan for a long question from Heredity and Evolution.

Answer:

Opening: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. Development: The mother contributes an X chromosome in the egg. Conclusion: Inherited traits result from genetic information passed through reproductive cells, such as natural eye colour. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Heredity and Evolution?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: If tallness T is dominant, $TT \times tt$ gives all Tt tall plants in F1. and The mother contributes an X chromosome in the egg.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.